

SolarLab vs. SCAPS-1D: A Drift-Diffusion Cross-Validation and Root-Cause Analysis of the Base-Model Discrepancy

Abstract

We compare two one-dimensional drift-diffusion (DD) solar-cell simulators — **SolarLab** (the Python `perovskite_sim` library under development in this repository) and **SCAPS-1D** (Burgelman/University of Gent, the de-facto reference) — on a partner “Base Model” perovskite stack: glass / spiro-OMeTAD hole-transport layer (HTL, 20 nm) / MAPbI₃-like absorber (800 nm, $E_g = 1.53$ eV) / TiO₂ electron-transport layer (ETL, 25 nm). Both codes are drift-diffusion solvers; neither is “experiment,” so we anchor absolute realism against the literature. At the base point, SolarLab reports $V_{oc} = 1.072$ V vs. SCAPS 1.1676 V (a -96 mV / -8% deficit), $J_{sc} = 25.73$ vs. 26.282 mA/cm² (-2%), FF = 85.6% vs. 86.99% (-1.4 pp), and PCE = 22.1% vs. 26.69% (-4.6 pp). Across ten device-parameter sweeps inherited from the SCAPS reference, only two levers close acceptably under a trend-first standard (conduction-band-offset 83% closure; PVK/ETL interface trap density 109%), and one shows an outright direction reversal (ETL donor doping). Through an adversarial three-lens review (physics-correctness, numerical-algorithm, data-consistency), we find that **the prior gap report’s central claim — a $37\times$ dark-saturation-current prefactor driven by quasi-Fermi-level dissipation across band offsets — is not supported by the conduction-band-offset sweep data**, which show the gap *widening* (not closing) as the offset is removed. The two robustly surviving root causes are: (i) the **frozen Poisson built-in-potential boundary** (`stack.V_bi = 1.30` V, hard-wired, while the band-derived V_{bi_eff} is computed but routed only to the sweep range), which severs the ETL-doping $\rightarrow V_{oc}$ lever and produces the direction reversal (3/3 lenses held); and (ii) the **PCE/FF self-consistency anomaly**, which is a *reporting artifact* (cross-run cell-splicing of figures-of-merit, not a solver defect; classification held, mechanism corrected). The -96 mV V_{oc} deficit itself remains *physical but contested in mechanism*: candidate drivers (hard-Dirichlet ohmic contacts, thermionic-emission flux capping, and the default interface-recombination model differing from SCAPS’s Pauwels–Vanhoutte formulation) each survived only one of three lenses. Critically, against experiment, **SolarLab’s 1.072 V sits near the measured-device median (1.05–1.13 V) while SCAPS’s 1.1676 V is at the published champion ceiling**; under the repository’s trend-over-absolutes house rule, we recommend exactly one cheap correctness fix (re-extract all four figures-of-merit from a single self-consistent J–V array to eliminate the PCE/FF splice) and explicitly recommend *against* the multi-week contact-BC / V_{bi} refactors needed to chase absolute V_{oc} parity.

1. Methodology

1.1 The two solvers

Both SolarLab and SCAPS-1D solve the standard one-dimensional semiconductor drift-diffusion system: Poisson’s equation coupled to electron and hole continuity equations, with Scharfetter–Gummel (SG) exponentially-fitted flux discretization [1, 5]. They differ in numerical strategy and several physical boundary treatments, summarized here and detailed in §3.

SCAPS-1D solves the DD set in the variables (ψ , E_{Fn} , E_{Fp}) — electrostatic potential and the two quasi-Fermi levels — using a **Gummel iteration with Newton–Raphson substeps** on a non-uniform finite-difference mesh with two coincident nodes at each metallurgical interface and adaptive refinement keyed to $\exp(q\psi/kT)$, $\exp(E_{Fn}/kT)$, and the R/G ratios (SCAPS Manual, Feb. 2016, pp. 39–41) [1, 6]. Carrier statistics are **Boltzmann by default**; Fermi–Dirac degeneracy and band-gap narrowing are explicitly not implemented (Manual p. 31). Interface recombination uses **Pauwels–Vanhoutte theory** (an SRH extension where both adjacent-layer band edges enter the effective rate), and interface *transport* is by **thermionic emission** with a thermal velocity equal to the smaller of the two adjacent layers, which always leaves a finite quasi-Fermi-level step at the interface even at zero band offset (Manual p. 30) [8].

SolarLab solves Poisson plus the two continuity equations by a **method-of-lines (MOL) pseudo-transient** scheme: space is discretized with exponentially-fitted SG finite-volume fluxes; Poisson is a single prefactored tridiagonal solve per right-hand-side (RHS) call (no inner Newton loop); and time is integrated to steady state with `scipy.integrate.solve_ivp(Radau)` (`perovskite-sim/perovskite_sim/solver/mol.py:1577–1622, 1741`). Convergence control is entirely Radau’s adaptive local-truncation-error machinery (`rtol = 1e-4`, `atol = 1e-6`), a `max_step` cap near flat-band, a `max_nfev` abort (`_JV_RADAU_MAX_NFEV = 100_000`), a BDF last-resort fallback

after bisections, and an `isfinite` RHS guard. Carrier statistics are **Boltzmann only**; n_i is supplied per-layer (in the validated path derived from the density of states via $n_i = \sqrt{(N_c \cdot N_v) \cdot \exp(-E_g/2kT)}$, matching the SCAPS convention; `loader.py:252`).

1.2 The partner Base Model stack

Layer	Material	Thickness	E _g (eV)	χ (eV)	Doping
HTL	spiro-OMeTAD	20 nm	(wide-gap)	2.40	p-type
Absorber	MAPbI ₃ -like	800 nm	1.53	3.94	intrinsic/lightly doped
ETL	TiO ₂	25 nm	(wide-gap)	4.10	n-type donor

The conduction-band offset at the PVK/ETL junction is $\Delta E_C = \chi_{PVK} - \chi_{ETL} = 3.94 - 4.10 = -0.16$ eV (a “cliff”). Interface defects are **neutral**, statistics are **Boltzmann**, and **tunnelling is OFF** in the partner configuration (default; the intra-band and band-to-band tunnelling checkboxes are unset, so transport over the cliff is pure thermionic emission) [1, 2]. The SolarLab mirror configuration is `perovskite-sim/configs/scaps_mirror_v2.yaml` (χ values at lines 89/124; $V_{bi} = 1.30$ at line 40; `mode: fast` at line 42, which keeps thermionic-emission capping ON and selective contacts OFF).

1.3 Data sources

- **SCAPS reference figures-of-merit and ten sweeps:** `perovskite-sim/tests/integration/scaps_reference.json` (base_model: V_{oc} 1.1676 V, J_{sc} 26.282 mA/cm², FF 86.99%, PCE 26.69%; plus 10 parameter sweeps with per-point V_{oc}).
- **Partner SCAPS report and parameter table:** `docs/superpowers/references/scaps_1d_simulation_report.pdf`, `docs/superpowers/references/scaps_1r_parameters.xlsx`.
- **SCAPS formulation:** `docs/SCAPS Manual february 2016.pdf` (111 pp.).
- **SolarLab sweep results (V_{oc} per point, SolarLab vs. SCAPS):** `outputs/scaps_full_off/summary.json` (interface projection OFF — the default path), `outputs/scaps_full_on/summary.json` (band-aware projection ON), `outputs/scaps_full_ifacestate/summary.json` (true Pauwels–Vanhoutte interface-plane state variant).
- **SolarLab full-FOM self-consistent artifact:** `perovskite-sim/outputs/scaps_validation/report.md` and `sweep_*.csv`.
- **Prior analysis (tested, not assumed):** `docs/partner/SolarLab_SCAPS_gap_analysis.md`.

1.4 Evaluation standard: trend-first

Per the repository house rule, **parity is judged by trend fidelity (direction + shape of each sweep) first, and absolutes second**. We do not propose multi-week refactors to chase absolute $V_{oc}/J_{sc}/FF/PCE$ parity. Closure is reported as a span ratio (SolarLab V_{oc} range ÷ SCAPS V_{oc} range over the sweep); we explicitly flag where this span-ratio metric masks a shape (curvature) mismatch (§4, N_t , PVK_ETL) or where the SolarLab sweep variable differs from the one in `report.md` (§4.1, CHI_ETL). Direction match is reported separately. Every quantitative claim below traces to one of the data sources in §1.3 or to a cited reference; literature values used only for realism-positioning are flagged as such.

2. Base-Model Comparison

2.1 Figures of merit

Metric	SolarLab	SCAPS-1D	Δ (SolarLab – SCAPS)
V_{oc} (V)	1.072	1.1676	-0.096 V (-8.2%) — primary discrepancy
J_{sc} (mA/cm ²)	25.73 †	26.282	-0.55 (-2.1%) — optical
FF (%)	85.6	86.99	-1.4 pp
PCE (%)	22.1	26.69	-4.6 pp

Sources: tests/integration/scaps_reference.json (SCAPS base_model); task FOM table and outputs/scaps_full_off/s (SolarLab base point, V_{oc} 1.0711 V at the true base $\Delta E_C = -0.16$ eV).

† The $J_{sc} = 25.73$ mA/cm² cell is a foreign value spliced from a different ($N_{grid} = 30$) run; SolarLab’s genuine self-consistent base-point J_{sc} is 23.96 mA/cm². See §2.2 and §5.3. It is retained here only because it is the as-quoted task tuple; every per-FOM analysis below uses 23.96.

2.2 PCE/FF self-consistency check (called out explicitly)

In any single, self-consistent J–V evaluation, the four FOMs are algebraically locked: with $FF \propto P_{mpp}/(V_{oc} \cdot J_{sc})$ and $PCE \propto P_{mpp}/(1000 \text{ W} \cdot \text{m}^{-2})$, it follows identically that $PCE = V_{oc} \cdot J_{sc} \cdot FF$ (perovskite-sim/perovskite-sim/experiment/120). The two reference tools obey this lock to within rounding:

- **SCAPS:** $1.1676 \times 26.282 \times 0.8699 = 26.69\% = \text{reported } 26.69\%$ (residual < 0.01 pp). □
- **SolarLab report.md:** $1.0905 \times 23.96 \times 0.8802 = 22.99\% = \text{reported } 22.99\%$ (residual < 0.01 pp). □

The **task-quoted SolarLab tuple does not obey the lock:** $1.072 \times 25.73 \times 0.856 = 23.61\%$, yet PCE is quoted as 22.1% — a **+1.51 pp residual that is mathematically impossible from a single compute_metrics call**. This is therefore a **reporting artifact** (a splice of FOM cells from two different runs), not a physics or solver defect. The adversarial review (§5) located the foreign cell precisely: SolarLab’s genuine self-consistent base-point rows carry $J_{sc} \approx 23.96$ mA/cm² (= 239.567 A/m² in the CSV artifacts), and a real, locked SolarLab point gives V_{oc} 1.0709 / J_{sc} 23.957 / FF 0.862 / **PCE 22.1%**. The **25.73 mA/cm² figure is the contaminant**: it originates from a different, later optimization run at $N_{grid} = 30$, where the peaked optical-generation profile inflates the trapezoidal J_{sc} integral by roughly +2.5 mA/cm² (docs/superpowers/specs/2026-05-29-e10-rootcause-optimization-workflow.md). The partner table glued that coarse-grid J_{sc} onto a fine-grid $V_{oc}/FF/PCE$ triple. **Consequence: the genuine SolarLab PCE is ≈ 22.1 – 22.3% , and the self-consistent value is not 23.6%** — quoting 23.6% would re-commit the very splice. (This corrects the prior gap report’s framing, which had assumed PCE was the stale cell.)

2.3 Three-way positioning against experiment

Neither solver is experiment; the question is which *absolute* is more representative of a fabricated MAPbI₃/spiro/TiO₂ n-i-p cell. The literature for $E_g \approx 1.53$ eV:

- **Radiative Shockley–Queisser V_{oc} ceiling:** ≈ 1.30 – 1.33 V at 1.53 eV; one detailed-balance study reports an SQ V_{oc} ceiling of ~ 1.25 V with a measured device reaching 1.18 V (94.4% of SQ) [12, 15]. The Auger limit lies within < 2 mV of the radiative limit, so SQ is the correct reference [14].
- **Champion measured V_{oc} :** A mesoporous-TiO₂/spiro champion delivers J_{sc} 24.6 mA/cm², **V_{oc} 1.16 V**, FF 0.73, PCE 20.8% [16] — this brackets SCAPS’s 1.1676 V as a *champion*, not a typical, value. Best-in-class low-loss stacks reach V_{oc} 1.21 V at 1.53 eV (certified 23.09% PCE , non-radiative V_{oc} loss ≈ 0.10 V), defining the physical ceiling for this exact gap [17].
- **Typical measured V_{oc} :** 1.05–1.13 V; single-crystal MAPbI₃ cells > 21% PCE sit at V_{oc} 1.0–1.1 V [18]. The typical non-radiative V_{oc} deficit is 150–280 mV below SQ [13].

Positioning: SCAPS 1.1676 V implies a non-radiative deficit of only ~ 130 – 160 mV (champion-grade interface passivation), consistent with the published champion ceiling but optimistic for a routine stack with a 20 nm spiro HTL and 25 nm TiO₂. SolarLab 1.072 V implies a ~ 230 – 260 mV deficit, squarely in the typical-device band. On PCE , **SCAPS’s 26.69% exceeds the certified single-junction MAPbI₃ record (~ 21 – 23%)** and lies above the SQ-derated practical ceiling, whereas SolarLab’s $\sim 22.1\%$ is realistic. *Conclusion: on absolutes, SolarLab is nearer the measured-device median; SCAPS is nearer the published champion. Neither is “wrong”; SolarLab’s absolute is the more conservative/representative of a typical fabricated cell.* (Literature V_{oc} values are well-established and cited; the SQ ceiling spread of 1.25–1.33 V reflects bandgap-definition variance and is flagged as a moderate-confidence band.)

3. SCAPS vs. SolarLab: Physical-Model and Numerical-Algorithm Comparison

Aspect	SolarLab (code file:line)	SCAPS-1D (manual page / DOI)
DD system / variables	Poisson + 2 continuity; carrier-density variables (n, p, ϕ). Poisson FV form $d/dx(\epsilon_0\epsilon_r d\phi/dx) = -\rho$, harmonic-mean face permittivity, Dirichlet BCs (poisson.py:111–182). Continuity $dn = (\nabla \cdot J_n)/q - R + G$; $dp = -(\nabla \cdot J_p)/q - R + G$ (continuity.py:212–213).	Poisson + 2 continuity in (ψ , E_{Fn} , E_{Fp}); $J_n = \mu_n n \cdot \nabla E_{Fn}$, $J_p = \mu_p p \cdot \nabla E_{Fp}$ (QFL-gradient form), Manual p. 39 eq. (14)–(15) [1].
Flux discretization	Exponentially-fitted Scharfetter–Gummel finite-volume: Bernoulli $B(x) = x/(e^x - 1)$ with small/large branches; flux $qD/h \cdot (B(\xi)n_{i+1}) - B(-\xi)n_i$ (fe_operators.py:9–81). True SG, not a plain FE gradient.	Scharfetter–Gummel-style flux on non-uniform FD mesh, 2 coincident nodes/interface (Manual p. 39–40) [5].
Solver / convergence	Method-of-lines pseudo-transient; Poisson = single LU-cached tridiagonal solve per RHS (no inner Newton); carriers advanced by Radau to steady state; rtol 1e-4, atol 1e-6, max_step cap, BDF fallback (mol.py:1577–1622, 1741).	Gummel scheme with Newton–Raphson substeps; adaptive mesh refinement on $\exp(q\psi/kT)$, R, G ratios ($f_{\max} \approx 1.60 \approx 12$ meV) (Manual p. 40) [1, 6].
Carrier statistics	Boltzmann only; n_i a per-layer scalar, $n_i^2 = n_i^2$ in every R term (parameters.py:17, 108–109). No Fermi–Dirac/degeneracy. n_i derived from DOS in the validated path (loader.py:252).	Boltzmann by default; Fermi–Dirac, degeneracy, band-gap narrowing not implemented (Manual p. 31). $n_i^2 = N_c N_v \exp(-E_g/kT)$.
Bulk SRH	$R = (np - n_i^2)/(\tau_p(n + n_i) + \tau_n(p + p_i))$ (recombination.py:5–10); n_i, p_i from trap-depth helper (device.py:25–26). Two PVK defects combined by parallel SRH $1/\tau_{\text{tot}} = \sum 1/\tau_i$ (loader.py:23–32).	SRH-through-defects; $\tau = 1/(\sigma N_t v_{th})$. Neutral defect $\rightarrow$ only τ enters the rate, not the space charge (Manual p. 24, 26) [9].
Auger / radiative	Auger ($C_n n + C_p p$)($np - n_i^2$) (recombination.py:20–25); radiative $B_{\text{rad}}(np - n_i^2)$ with per-RHS Yablonoitch P_{esc} photon-recycling reabsorption (recombination.py:13–17).	SRH + radiative + Auger bulk channels; no per-RHS photon-recycling reabsorption (Manual recombination sections).
Interface SRH (Pauwels–Vanhoutte)	Default: velocity-form face-SRH $R = (np - n_i^2)/((n + n_i)/v_p + (p + p_i)/v_n)$, $v = \sigma v_{th} N_t$ (recombination.py:28–44; loader.py:196–202), sampled at bulk-interior cross-carrier nodes (n from transport side, p from absorber side) with $n_i^2_{\text{eff}} = n_R \text{eq} p_L \text{eq}$ and an $R \geq 0$ clamp (mol.py:1188–1234). A true Pauwels–Vanhoutte interface-plane-state model (n_{1s} , p_{1s} , n_{2s} , p_{2s} + TE coupling + two-sided SRH) exists but is env-gated OFF (interface_plane.py:49–268).	Single Pauwels–Vanhoutte interface-state SRH: electrons in layer-1 CB and holes in layer-2 VB recombine; both band edges set the effective n, p in the SRH denominator (Manual p. 30; Pauwels & Vanhoutte, J. Phys. D 11 (1978) 649) [8].
Contact BC	Default hard Dirichlet ohmic pin: $dn[0] = dn[-1] = 0$, carriers held at doping-equilibrium $n_R = N_D$, $p_R = n_i^2/N_D$ ($S \rightarrow \infty$ ohmic) (continuity.py:216–223; mol.py:874–888). Optional finite-S Robin path $J = \pm qS(n - n_{\text{eq}})$ exists but is supplied by no shipped preset (contacts.py:62–134).	Metal work-function Φ_m for majority carriers, or “flat bands” (Φ_m solved for flatband); shallow doping in eq. (1)–(3); wavelength-dependent reflection/transmission filter at the contact (Manual p. 9–10) [1].
Poisson built-in potential	Frozen manual scalar $\text{stack.V}_{bi} = 1.30$ V used at the boundary $\phi_{\text{right}} = \text{stack.V}_{bi} - V_{\text{app}}$ (mol.py:1334, 1731; config line 40). Band-derived $V_{bi_eff} = \text{compute_V}_{bi}()$ (Fermi-level difference, tracks χ_{ETL} and N_D) is computed but routed only to the V_{max} sweep range (mol.py:895, 1054; device.py:186).	Built-in potential set self-consistently by the work-function/flat-band contact electrostatics; moves with band edges and doping (Manual p. 9–10) [1].
Band-offset transport / tunnelling	Vacuum level continuous ($\phi_n = \phi + \chi$, $\phi_p = \phi + \chi + E_g$); at $ \Delta E_C $ or $ \Delta E_V > 0.05$ eV the SG flux is capped to the Richardson–Dushman thermionic-emission bound (continuity.py:144–181). TE applied as a flux ceiling on top of SG , not as the interface BC.	Thermionic emission as the interface transport BC (always a finite QFL step); intra-band and band-to-band tunnelling are optional checkboxes, OFF in the partner model (Manual p. 30–31; Burgelman & Marlein, TSF 515 (2007) 6276) [2].
Optics	TMM front reflection over glass $\rightarrow$ spiro $\rightarrow$ PVK; wavelength-resolved $R(\lambda)$; glass index-matching recovers most of the bare-stack Fresnel loss (optics.py; config lines 47–51). Finite-thickness absorber transmission and a dropped multi-pass geometric series mean net optical loss exceeds pure front R (optics.py:482–488).	Internal generation $g(x)$ from AM1.5G (1000 W/m ²); front reflection = 0 unless a reflection/filter .spe is supplied (none in the partner model) $\rightarrow$ near-ideal absorption (Manual p. 10, 36–37).
FOM extraction	$J_{sc} = \text{interp}(0, V, J)$; V_{oc} by linear interpolation at the first J sign-crossing; $FF = P_{mpp}/(V_{oc} J_{sc})$; $PCE = P_{mpp}/1000$ (jv_sweep.py:105–120). Terminal J for the sweep uses the contact face $J_{\text{faces}}[0]$ (carries displacement transients), not the charge-conserving median (jv_sweep.py _compute_current vs. _compute_current_ss:377).	$V_{oc} J_{sc} FF$ self-consistent (PCE 26.69 = 1.1676·26.282·0.8699).

Key model-level divergences feeding the base-point discrepancy: (1) the **frozen V_{bi} boundary** (SolarLab) vs. the doping-/band-tracking flatband contact (SCAPS); (2) the **TE flux-cap-on-top-of-SG** (SolarLab) vs. **TE-as-the-BC** (SCAPS); (3) the **default bulk-projected cross-carrier velocity-SRH interface model** (SolarLab)

vs. **single-plane Pauwels–Vanhoutte both-band-edge SRH** (SCAPS); (4) **hard-Dirichlet ohmic contacts** (SolarLab default) vs. flatband work-function contacts (SCAPS); (5) **TMM front reflection + finite-absorber transmission** (SolarLab) vs. **R = 0 ideal optics** (SCAPS).

4. Step-by-Step Sweep Analysis (10 sweeps)

The SolarLab side uses the default path (interface projection OFF, outputs/scaps_full_off/summary.json); ON figures (scaps_full_on) are noted where they matter. “Closure” = SolarLab V_{oc} span $\div$ SCAPS V_{oc} span; “Dir” = direction match; “Base gap” = SolarLab – SCAPS V_{oc} at the sweep’s base point.

#	Sweep (lever)	SCAPS span (mV)	SolarLab span (mV)	Closure	Dir	Base gap (mV)	Group
1	CHI_ETL (CBO)	918.0	761.7 ‡	83.0%	□	–96.5	matched (slope)
2	Nt_PVK_ETL (iface N_t)	226.7	246.2	108.6%	□	–338.7	divergent (shape/abs)
3	Nd_ETL (ETL donor)	99.6	29.7	29.8%	×	–96.5	divergent (direction)
4	Nt_HTL_PVK (iface N_t)	5.2	85.0	1633%	×	–69.3	divergent (over-sensitive)
5	Nt_C_PVK (bulk N_t, CB)	38.6	0.1	0.2%	□	–95.9	divergent (dead)
6	Nt_V_PVK (bulk N_t, VB)	10.8	0.1	0.6%	□	–95.8	divergent (dead)
7	Et_PVK_ETL (iface trap E)	34.5	0.0	0.0%	×	–96.5	divergent (dead)
8	Et_C_PVK (bulk E_t, CB)	0.4	3.4	787%	□	–96.0	divergent (over-sensitive)
9	Et_V_PVK (bulk E_t, VB)	0.4	3.4	783%	□	–95.8	divergent (over-sensitive)
10	Et_HTL_PVK (iface trap E)	0.0	0.0	flat	flat	–96.5	matched (both flat)

‡ The 761.7 mV SolarLab span is the absolute- χ -drag sweep variable in summary.json (the ETL affinity itself is dragged across the full range). It is *not* the same lever as the 18 mV span quoted in report.md, which sweeps the band-aligned ΔE_C offset (FF stays ~ 0.83 and the junction keeps working). The two SolarLab numbers are the same code on two different sweep-variable definitions; the reconciliation is detailed in §4.1 and §5.6.

4.1 Matched / acceptable

Sweep 1 — CHI_ETL (conduction-band offset). SCAPS’s dominant V_{oc} lever (918 mV range over 14 points). SolarLab reproduces 83% of the span and the direction is correct, the **best trend match**. *But the gap is not constant:* it is –7 mV in the deep-cliff regime ($\Delta E_C = -1.0$ to -0.5 eV, where the real cliff barrier limits V_{oc} in both tools and they agree), –96.5 mV at the base (-0.16 eV), and grows to –163 to –166 mV at flat-band/spike ($\Delta E_C \geq 0$). SolarLab’s V_{oc} saturates near 1.085 V for $\Delta E_C \geq -0.1$ eV (6.0 mV spread) while SCAPS keeps climbing to 1.25 V (33 mV further). Projection ON raises slope closure to 92% but lowers base V_{oc} by ~ 19 mV. **A note on the two SolarLab spans:** the summary.json value used in the table above is **761.7 mV** because that sweep drags the absolute electron affinity χ_{ETL} across its full range (which eventually destroys the junction at extreme negative offset); report.md reports an **18 mV** span for the *same* code because it sweeps the band-aligned ΔE_C with the rest of the band structure shifted self-consistently, so the device keeps working and V_{oc} moves little. Both are correct for their respective sweep-variable definitions; they are not in conflict. **Inline root cause (contested):** the saturation signature is consistent with the frozen- V_{bi} boundary failing to track χ_{ETL} , and/or the TE flux cap dissipating QFL in the spike regime — the two mechanisms could not be cleanly separated (see §5).

Sweep 10 — HTL/PVK interface trap energy (Et_HTL_PVK). SCAPS itself is dead here (0.0 mV) and SolarLab is dead (0.0 mV). Both flat $\rightarrow$ trivially “matched.” Not diagnostic.

4.2 Divergent

Sweep 3 — Nd_ETL (ETL donor doping): DIRECTION REVERSAL. SCAPS rises monotonically +99.6 mV with doping (sub-ideal ~14 mV/decade); SolarLab nets **-11 mV** (closure 29.8%, dir_match = false). This is the cleanest, best-supported root cause in the study (3/3 lenses held). **Inline root cause:** SolarLab freezes the Poisson boundary at `stack.V_bi = 1.30 V` (`mol.py:1334, 1731`), so raising N_D cannot lift the built-in potential. The chi-/doping-aware `V_bi_eff = compute_V_bi()` (`device.py:103-134`) *does* track N_D but is routed only to the sweep range, never the boundary. The only residual N_D channel is the ohmic pin $n_R = N_D$, $p_R = n_i^2/N_D$ (`mol.py:874-888`): raising N_D *lowers* the pinned minority-hole density, slightly *suppressing* QFL splitting → wrong sign. The raw rows show the predicted non-monotonic signature: V_{oc} falls 1.0954 → 1.0656 V ($1e13$ → $1e16$ cm⁻³, pin tightening) then drifts up only to 1.0844 V ($1e20$) from second-order electrostatics. The very-low- N_D rows ($\leq 1e12$ cm⁻³) fail to converge ($V_{oc} = 0$), an extraction failure separate from the V_{bi} mechanism. *Caveat surfaced in review:* the SCAPS lever's dominant low-doping gain is actually FF/extraction recovery (~7.7 mV/dec, far below the 59.5 mV/dec ideal flatband slope), with a genuine V_{bi} lift only in the top ~2 decades — SolarLab severs both the V_{bi} lever and the FF-recovery lever. Classification: **both (numerical boundary-condition severance + physical contact choice)**; confidence high; survived 3/3.

Sweep 2 — Nt_PVK_ETL (PVK/ETL interface trap density): SPAN MATCHES, SHAPE DOES NOT. The headline 108.6% “closure” is a **span-ratio artifact**. Per-decade, the two curves have *opposite curvature*: SCAPS is plateau-then-knee (-1 to -12 mV/dec below $1e11$ cm⁻², then -68 to -79 mV/dec at $1e12$ - $1e13$), while SolarLab is **front-loaded** (-76 mV/dec from the first decade, saturating to -9 mV/dec exactly where SCAPS is steepest; SolarLab also returns NaN at $N_t = 1e15$). The base gap is -338.7 mV — the worst of any sweep — so the absolute is badly over-recombining. **Inline root cause:** SolarLab's velocity-form face-SRH puts N_t only in the denominator via $v = \sigma \cdot v_{th} \cdot N_t$ (`recombination.py:28-44`), giving a velocity-limited → density-limited crossover (front-loaded then saturating), which is the wrong shape for an interface-state SRH whose recombination-active occupancy turns on near a threshold N_t ; SCAPS's single-plane Pauwels-Vanhoutte rate stays plateaued until the interface density competes with bulk extraction (the late knee). The over-recombination at low N_t reflects sampling near-majority bulk-interior densities (`mol.py:1188-1234`) so $(np - n_i^2)$ is large even at negligible N_t . The true interface-plane-state path (`interface_plane.py`) is mis-calibrated (the ifacestate variant collapses sensitivity to 2%). Classification: **both (physical interface-SRH model + numerical sampling location)**; the prior framing of “slope reproduced, only absolute wrong” did **not** survive — the shape is the part that fails.

Sweep 4 — Nt_HTL_PVK (HTL/PVK interface trap density): OVER-SENSITIVE. SCAPS moves only 5.2 mV; SolarLab moves 85 mV (1633% closure, dir_match = false). Same velocity-SRH model as Sweep 2, here far too responsive at the HTL/PVK plane — consistent with the cross-carrier interior-node sampling over-weighting this interface.

Sweeps 5, 6 — Nt_C/V_PVK (PVK bulk trap density): DEAD. SCAPS moves 38.6 / 10.8 mV; SolarLab moves 0.1 / 0.1 mV (closure 0.2% / 0.6%). The SRH rate form and the parallel-defect combination are coded correctly (`recombination.py:5-10`; `loader.py:23-32`), so the term is right but not rate-limiting at the operating point. **Inline root cause (contested):** the bulk channel is sub-dominant. Adversarial review showed the cleaner physics is **carrier-density suppression** — SolarLab's absorber n-p at V_{oc} is low (the same root as the -96 mV base offset: lower QFL splitting → small np), so $(np - n_i^2)$ is tiny and the N_t -linear prefactor is invisible. The “transport-ceiling-masks-bulk” framing did *not* survive: a single series ceiling would suppress the E_t lever equally, yet SolarLab is N_t -dead but E_t -hypersensitive (3.4 mV vs. SCAPS 0.4 mV, ~8×; see Sweeps 8-9), which demands that n_1, p_1 still matter in the denominator — i.e. carriers are starved, not masked. Both tools are *also* base-flat below $1e13$ cm⁻³ (100 μs lifetime), so much of SCAPS's 38.6 mV comes from the single $1e14$ → $1e15$ decade. Classification: physical-model; the dead bulk response is a *symptom* of the V_{oc} -suppression root, not an independent bug.

Sweep 7 — Et_PVK_ETL (interface trap energy): DEAD + DIRECTION MISMATCH. SCAPS moves 34.5 mV with trap position; SolarLab moves 0.0 mV. The interface-SRH denominator's n_1, p_1 are insensitive to trap position in the default path — a real but secondary defect, distinct from the N_t -magnitude question.

Sweeps 8, 9 — Et_C/V_PVK (PVK bulk trap energy): SolarLab MILDLY OVER-SENSITIVE. SCAPS is essentially dead (0.4 / 0.4 mV) while SolarLab moves **3.4 / 3.4 mV** — an ~8× over-sensitivity, so the span-ratio closure metric reads **787% / 783%** (it is not “flat” on the SolarLab side). The “both flat” description is true only for SCAPS. This is the same E_t -hypersensitivity that §5.5's verifier uses (3.4 vs. 0.4 mV, “ E_t -hypersensitive”) as evidence that SolarLab's absorber carriers are *starved* rather than masked: because $n_1, p_1 \sim n_i \cdot \exp(\pm E_t/kT)$ still matter in the SRH denominator, a small trap-energy shift moves V_{oc} , which can only happen when n-p at V_{oc} is low. Direction

matches and the absolute movement is tiny (3.4 mV), so this is a low-severity divergence — but it is *not* a clean match.

5. Root-Cause Findings

Each finding carries its classification, mechanism, evidence, rejected alternatives, confidence, cost of forcing agreement, and — critically — its **adversarial survival** across three lenses (physics-correctness, numerical-algorithm, data-consistency). Causes that did not survive a majority of lenses are marked **TENTATIVE**.

5.1 V_{oc} -96 mV primary deficit — TENTATIVE (mechanism contested; 1/3 lenses held)

- **Classification:** physical-model (boundary/transport), with a numerical reporting cross-check that is separable.
- **Proposed mechanism (as advanced):** SolarLab’s hard-Dirichlet ohmic outer contacts pin the minority carrier at doping-equilibrium (infinite surface-recombination velocity), injecting a parasitic contact-recombination current absent in SCAPS’s finite-velocity flatband contacts; the implied dark-saturation prefactor ratio $J_0,SL/J_0,SCAPS \approx 33\text{--}42\times$ translates to $V_T \cdot \ln(\approx 33\text{--}42) = 90\text{--}96\text{ mV} \approx$ the observed deficit (`continuity.py:216-223`; `mol.py:874-892`).
- **Evidence for:** The J_0 ratio and its $V_T \cdot \ln$ translation are arithmetically verified ($J_0,SL \approx 2.5e\text{--}20$, $J_0,SCAPS \approx 6.2e\text{--}22\text{ A/cm}^2$; $V_T \cdot \ln(41.7) = 96.5\text{ mV}$ = the exact base gap). The deficit is genuinely physical (reproduced by the validated sweeps, FF healthy at 0.878 in the self-consistent run, so not a terminal-face artifact). V_{bi} headroom (band-derived 1.294 V $\approx$ manual 1.30 V) and n_i (Boltzmann-consistent) are excluded as drivers. The TE cap does *not* bind on the forward light current ($v_{TE}/v_{diff} \approx 1045$).
- **Why it is TENTATIVE (refutations that held):**
 1. *Physics lens (REFUTED):* A hard-Dirichlet ohmic pin is the **same** $S \rightarrow \infty$ minority condition SCAPS’s flatband ohmic contact uses, so by itself it cannot produce a code-to-code difference. The implied J_0 for both tools is consistent with a **bulk-SRH-limited** diode (both operating points sit between the radiative floor $J_0 \approx 2.5e\text{--}23$ and the SRH level), so the $33\text{--}42\times$ is the ratio of two SRH-dominated diodes — better explained by the two codes computing **different interface SRH rates** (SolarLab’s default bulk-projected cross-carrier SRH vs. SCAPS Pauwels–Vanhoutte both-band-edge SRH), which is untested here.
 2. *Data-consistency lens (REFUTED):* Reading the full 14-point CHI_ETL rows shows the gap is **monotone in CBO** (−7 mV at the deep cliff $\rightarrow$ −163 mV at flat-band), not a constant contact plateau. Agreement to −7 mV at the deep cliff cannot arise from an $S \rightarrow \infty$ sink that is unchanged across the sweep. Projection ON (a pure interface-recombination lever) moves base V_{oc} by −19 mV, which a metal-contact sink should be insensitive to. Both point to a **band-offset / interface-transport** mechanism, not a contact sink. The “ $33\times$ ” figure is also numerically loose; the exact ratio is $41.7\times$.
 3. *Numerical lens (HELD):* The deficit is not a discretization/FOM artifact — V_{oc} is face-choice-insensitive at the crossing, the PCE/FF inconsistency is V_{oc} -orthogonal, and the TE cap does not bind. The base deficit is better quoted as **−77 to −96 mV** (the self-consistent `report.md` run gives −77 mV: 1.0905 vs. 1.1676), with ~19 mV run-to-run spread.
- **Net status:** The *symptom* (an elevated effective J_0 of $\approx 40\times \rightarrow \approx 96\text{ mV}$) and the *exonerations* (V_{bi} , n_i , TE-cap, FOM-extraction as drivers of the base V_{oc}) are robust. The *attribution* to a contact-specific minority sink is **not established and probably wrong**; the better-supported (but not independently confirmed) origin is the **difference between SolarLab’s default interface-SRH model and SCAPS’s Pauwels–Vanhoutte interface-state + thermionic-emission interface BC**. The prior gap report’s “ $37\times J_0$ from QFL dissipated ~135 mV across band offsets” is **falsified at the base point by the CBO-sweep sign** (gap widens, not closes, as the offset is removed).
- **Confidence:** medium that the cause lies in interface SRH/transport; low that it is the contact BC specifically.
- **Missing experiment (the one that would settle it):** a controlled finite-S contact re-run and a per-channel $R(x)$ recombination-budget dump at V_{oc} ; neither was executed (budget-limited).
- **Cost of forcing agreement:** switching to finite-S selective/flatband contacts or routing V_{bi_eff} into the boundary would re-baseline every regression, change FF/ J_{sc} on all presets, risk Radau stiffness from the Robin time constant dx/S , and tune SolarLab *off* the experimental median (1.05–1.13 V) *toward* SCAPS’s champion ceiling (1.168 V) — a fidelity loss against literature. **Not recommended.**

5.2 Nd_ETL direction reversal — ROBUST (3/3 lenses held)

- **Classification:** both (numerical boundary-condition severance, compounded by the physical contact choice).
- **Mechanism:** The Poisson right BC is hard-wired to `stack.V_bi = 1.30 V` (`mol.py:1334, 1731, 1759`); the N_D -dependent `V_bi_eff = compute_V_bi()` (`mol.py:895`, via `_fermi_level` reading N_D) flows only to $V_{\max}$, never the boundary. With V_{bi} frozen, raising N_D cannot lift the built-in potential; the only residual channel is the ohmic pin ($p_R = n_i^2/N_D$ as $N_D \rightarrow \infty$), which slightly suppresses QFL splitting $\rightarrow$ wrong sign.
- **Evidence:** Direction data `sl_dir = -0.0109`, `sc_dir = +0.0996`, closure 29.8%, `dir_match = false` (`outputs/scaps_full_off/summary.json`). Predicted non-monotonic SolarLab signature (fall 1.0954 $\rightarrow$ 1.0656, then rise to 1.0844) confirmed in raw rows. Projection ON preserves the wrong sign (`sl_dir = -0.0111`) — interface recombination is downstream of, and cannot restore, the severed lever. All three lenses independently confirmed the code path and the fall-then-rise signature; the ON/OFF discriminator passed (an interface-recombination root cause would have flipped direction under ON; it does not).
- **Rejected alternatives:** the -96 mV base offset (flat across the sweep, cannot cause a slope-sign error); the TE cap (ΔE_C fixed under N_D); n_i (per-layer fixed, shifts absolute not derivative sign). All correctly excluded.
- **Confidence:** high. One nuance surfaced in review: SCAPS's headline Nd_ETL benefit is largely FF/extraction recovery at low doping (7.7 mV/dec $\square$ 59.5 mV/dec ideal), not a clean flatband V_{bi} lift; SolarLab severs both levers. The very-low- N_D ($\leq 1e12 \text{ cm}^{-3}$) SolarLab $V_{\text{oc}} = 0$ rows are a convergence/extraction failure, not part of the V_{bi} mechanism.
- **Cost of forcing agreement:** routing `V_bi_eff` (or a flatband Robin contact) into the boundary recovers the $+N_D$ direction but shifts the base-point V_{bi} off 1.30 V, perturbs the calibrated base offset and every other sweep's base V_{oc} , breaks the documented IonMonger-convention invariant and the 17/17 guard/regression baselines, and reintroduces the contact time-constant the hard pin avoids. **Real defect, worth flagging; the BC rework is the multi-week refactor the house rule warns against — not recommended for parity.**

5.3 PCE/FF self-consistency anomaly — ROBUST classification, CORRECTED mechanism

- **Classification: reporting-artifact** (not physics, not solver). This top-level label held across all three lenses.
- **Mechanism (corrected):** FF and PCE both derive from the same P_{mpp} scalar (`jv_sweep.py:118-120`), so $\text{PCE} \square V_{\text{oc}} \cdot J_{\text{sc}} \cdot \text{FF}$ in any single run. The task tuple violates this by +1.51 pp, definitionally a multi-source pairing. **The foreign cell is J_{sc} (25.73 mA/cm²), not PCE.** SolarLab's genuine self-consistent base rows carry $J_{\text{sc}} = 23.957 \text{ mA/cm}^2$ and $\text{PCE} \approx 22.1\%$ (locked: $1.0709 \cdot 23.957 \cdot 0.862 = 22.1\%$); the 25.73 figure comes from a separate $N_{\text{grid}} = 30$ optimization run whose peaked-G trapezoidal integral inflates J_{sc} by $\sim +2.5 \text{ mA/cm}^2$. The prior gap report's framing (PCE was the stale cell; "true" $\text{PCE} \approx 23.6\%$) is **wrong** — re-quoting 23.6% would re-commit the splice.
- **Evidence:** Algebraic lock verified in source; `report.md` (22.99% locked) and SCAPS (26.69% locked) both self-consistent to < 0.01 pp; the contaminated tuple appears only in partner-facing prose (`docs/partner/SolarLab_SCAPS_gap_analysis.md`), never in any pipeline output; the J_{sc} back-solve ($\text{PCE } 22.1\%$ needs $J_{\text{sc}} \approx 24.08$ at fixed V_{oc}/FF , matching the real 23.96 within rounding) pinpoints J_{sc} as the spliced cell.
- **Rejected alternatives:** an FF physics defect (would move FF and PCE together, preserving the lock); the terminal-face $J_{\text{faces}}[0]$ noise (real sweep CSVs are self-consistent to 0.000 pp using that very path, so it is not the contaminant here); the -96 mV V_{oc} physics (orthogonal — PCE and FF are algebraically locked).
- **Confidence:** high on classification and on J_{sc} -as-contaminant.
- **Cost of forcing agreement: near-zero.** Re-extract all four FOMs from one J-V array in a single `compute_metrics` call and quote that self-consistent tuple. This is the one cheap correctness fix recommended below.

5.4 $J_{\text{sc}} -2\%$ (front optics) — partially TENTATIVE

- **Classification:** expected-physical-residual (optical assumption mismatch, not a numerical bug). This held.
- **Mechanism (corrected):** SolarLab computes a real front reflectance via TMM (glass/spiro/PVK); SCAPS uses $R = 0$ by default. *But the $-2.10\% = (1-R)$ closure in the prior framing is built on the spliced $J_{\text{sc}} =$*

25.73. The genuine validated SolarLab J_{sc} is **23.96 mA/cm²** (`report.md`; sweep CSVs), so the real deficit vs. SCAPS 26.282 is **-2.33 mA/cm² = -8.8%**, not -2%. A ~5% glass-stack front reflection (config lines 47–51) cannot account for 8.8%; the residual ~3–4 pp is **absorption-side** — finite-thickness 800 nm transmission and a dropped multi-pass geometric series in `optics.py`:482–488, plus parasitic HTL/ETL/glass absorption — which SCAPS’s near-unity internal absorption recovers.

- **Status:** the *direction and classification* (SolarLab < SCAPS, optical, decoupled from the V_{oc} transport gap) survive; the *magnitude and single-cause “front reflection” attribution* did **not** (3/3 refuted). Possible numerical contributors to the absorption residual (AM1.5G spectral quadrature resolution, trapezoidal optical-depth integration on the tanh grid, the `_inv2x2` determinant-guard clamp) were flagged but not excluded.
- **Confidence:** high that the residual is optical and V_{oc} -orthogonal; low that it is pure front reflection.
- **Cost of forcing agreement:** disabling TMM reflection ($R = 0$) would discard a physically real loss and make SolarLab’s J_{sc} less representative of fabricated cells (champion MAPbI₃ $J_{sc} \approx 24.6$ mA/cm²; SolarLab 23.96 already realistic). **Not recommended** — but the J_{sc} table cell should be quoted as 23.96, not 25.73 (see §5.3).

5.5 Bulk-trap insensitivity (N_t C/V_PVK dead) — TENTATIVE mechanism (1/3 held)

- **Classification:** physical-model.
- **Status:** The SRH form and wiring are correct (held across lenses); the dead N_t response is real. The proposed “transport ceiling masks a sub-dominant parallel bulk channel” mechanism was **REFUTED on physics and numerical lenses**: it fails the high- N_t limit (SolarLab shows zero bend where SCAPS reaches 30 mV/dec) and the N_t -dead / E_t -hypersensitive asymmetry (N_t closure 0.2% but E_t span 3.4 mV vs. SCAPS 0.4 mV, $\sim 8\times$ — see Sweeps 8–9), which together indicate **carrier-density suppression** (small np at V_{oc}) rather than series masking. The data-consistency lens *held* (the attenuated-but-faithful, monotonically-un-masking SolarLab response rules out a wiring bug). **Net: dead bulk response is a symptom of the same V_{oc} -suppression root as §5.1, not an independent defect.**
- **Confidence:** medium that it is a symptom; the precise mechanism shares the §5.1 uncertainty.

5.6 CHI_ETL absolute saturation (frozen V_{bi} vs. TE cap) — TENTATIVE (1/3 held)

- **Classification:** both (numerical BC + physical TE cap), primary contested.
- **Status:** The frozen- V_{bi} code asymmetry is real and confirmed (numerical lens HELD; it is a *documented* IonMonger-convention choice, not an accidental bug). But the *electrostatic-cap mechanism* was **REFUTED on physics and data lenses**: SolarLab saturates at ~ 1.085 V, ~ 215 mV **below** the frozen 1.30 V boundary, so V_{bi} is not the binding constraint; and projection ON moves the saturation ceiling while the Poisson boundary is byte-identical, proving the ceiling is set by **interface recombination**, not the BC. The physics lens argues the surviving chi-coupled interior term is the **TE flux cap** (the demoted “compounding” factor). **Net: the saturation is recombination/QFL-limited, not electrostatically clamped; the frozen- V_{bi} bug is real but inert for V_{oc} at the base point (its live consequence is the N_d ETL reversal, §5.2).** Note that the **761.7 mV SolarLab span** quoted for this sweep in `summary.json` is the absolute- χ -drag variable; `report.md`’s **18 mV** span is the band-aligned ΔE_C variable on the same code (see §4.1).

6. Synthesis and Remediation

6.1 Genuine physical-model differences (real, by design or by formulation)

1. **Interface-recombination model.** SolarLab’s default velocity-form bulk-projected cross-carrier SRH (`recombination.py`:28–44; `mol.py`:1188–1234) is *not* SCAPS’s single-plane Pauwels–Vanhoutte both-band-edge SRH [8]. This drives the N_t _PVK_ETL shape mismatch (§4.2/Sweep 2), the N_t _HTL_PVK over-sensitivity (Sweep 4), the E_t _PVK_ETL deadness (Sweep 7), and is the **most likely true origin of the -96 mV base deficit** (§5.1). A true Pauwels–Vanhoutte path exists (`interface_plane.py`) but is mis-calibrated/off.
2. **Thermionic-emission treatment.** SolarLab caps the SG flux at the Richardson bound for $|\Delta E_C| > 0.05$ eV (`continuity.py`:144–181); SCAPS uses TE as the interface BC (Manual p. 30) [8]. This shapes the CHI_ETL spike-regime gap (§5.6) and is the surviving candidate for that saturation.

3. **Contact statistics.** SolarLab default = hard-Dirichlet ohmic ($S \rightarrow \infty$); SCAPS = flatband work-function contacts (Manual p. 9–10). Same $S \rightarrow \infty$ minority limit, so *not* by itself a code-to-code V_{oc} driver (§5.1, refuted) — but the frozen- V_{bi} coupling makes it the live cause of the Nd_{ETL} reversal (§5.2).
4. **Optics.** SolarLab TMM front reflection + finite-absorber transmission vs. SCAPS $R = 0$ (§5.4). A real, bounded physical residual.
5. **Carrier-density suppression at V_{oc} .** The low absorber n_p that masks the bulk-trap lever (§5.5) is a *consequence* of (1)/(2), not an independent difference.

6.2 Numerical-algorithm artifacts (true code-level issues)

1. **Frozen Poisson V_{bi} boundary** (`mol.py:1334, 1731`): the band-/doping-derived V_{bi_eff} is computed but never used at the boundary. **Live defect** — produces the Nd_{ETL} direction reversal (§5.2). Documented IonMonger convention, so a deliberate design liability rather than an accidental error, but a liability with a real cost.
2. **PCE/FF FOM splice** (reporting layer, not solver): cross-run cell-mixing in partner-facing tables (§5.3). **The only cheap-to-fix correctness issue.**
3. **Terminal-face current convention** (`jv_sweep.py _compute_current = J_faces[0]`): carries displacement transients vs. the charge-conserving median used by the SS probe. *Not* the V_{oc} driver and *not* the PCE-splice contaminant (verified), but a latent FF/P_mpp noise source worth standardizing on the median.
4. **Low- N_D convergence failures** ($Nd_{ETL} \leq 1e12 \text{ cm}^{-3} \rightarrow V_{oc} = 0$): an extraction/convergence gap at the lever extreme, orthogonal to the direction reversal.

6.3 Expected / bounded residuals (do not chase)

- **J_{sc} –2% (table) / –8.8% (true):** optical, V_{oc} -orthogonal, partially absorption-side; SolarLab’s 23.96 mA/cm² is *more* realistic than SCAPS’s $R = 0$ idealization (§5.4).
- **V_{oc} –96 mV:** physical; SolarLab sits at the experimental median, SCAPS at the champion ceiling (§2.3). Under trend-first, document — do not tune.
- **FF –1.4 pp:** consequent on V_{oc} and the terminal-face convention; not independently meaningful once the FOM splice is corrected.

6.4 Prioritised remediation (honest)

Priority	Action	Class	Recommendation
P0 (do now)	Re-extract all four FOMs from a single J–V array in one <code>compute_metrics</code> call; quote SolarLab’s self-consistent base tuple (V_{oc} 1.071 / J_{sc} 23.96 / FF ~0.862 / PCE 22.1%). Stop quoting J_{sc} 25.73 (foreign N_{grid} =30 cell).	Reporting fix	DO — near-zero risk, removes the –4.6 pp PCE artifact and the +1.51 pp self-consistency violation.
P1 (cheap, optional)	Standardize the swept J–V on the charge-conserving median face (<code>_compute_current_ss</code>) instead of <code>J_faces[0]</code> ; regenerate any text citing 85.6%/22.1% as a matched pair.	Numerical hygiene	DO if convenient — small FF/P_mpp denoising, no physics change.
P1 (documentation)	Document the frozen- V_{bi} boundary as the cause of the Nd_{ETL} direction reversal; note V_{bi_eff} is intentionally routed only to V_{max} (IonMonger convention).	Documentation	DO — honest disclosure; do not silently hot-fix.
P2 (investigate, not fix)	Run the missing controlled experiments for the –96 mV: a finite-S contact re-run and a per-channel $R(x)$ budget dump at V_{oc} , to settle interface-SRH-rate vs. contact-BC attribution.	Investigation	OPTIONAL — diagnostic only.

Priority	Action	Class	Recommendation
NOT RECOMMENDED	Route V_{bi_eff} into the Poisson boundary; switch defaults to finite-S flatband contacts; recalibrate the true Pauwels–Vanhoutte interface-plane model as default; disable TMM front reflection.	Costly refactor	DO NOT — each re-baselines the 17/17 guard suite and every sweep’s base point, risks Radau stiffness, and (for V_{oc}) tunes SolarLab away from the experimental median toward SCAPS’s optimistic champion ceiling. Violates the trend-over-absolutes house rule for an absolute-parity gain that is not physically more correct.

6.5 Bottom line

SolarLab reproduces SCAPS’s **dominant V_{oc} lever (conduction-band offset) at 83% span fidelity with correct direction**, which is the trend that matters most. Its absolute V_{oc} (1.072 V) and PCE (~22.1%) are *more representative of fabricated MAPbI₃ devices* than SCAPS’s champion-ceiling absolutes. The genuine defects are: one **reporting splice** (P0, fix now — it is the source of the headline –4.6 pp PCE gap and the J_{sc} table inconsistency) and one **frozen-boundary direction reversal** on ETL doping (document, do not refactor). The prior report’s central physical claim — a $37\times$ J_0 from QFL dissipated across band offsets — does **not survive** the conduction-band-offset sweep data and should be retracted in favour of the interface-SRH-model difference as the better-supported (though not yet independently confirmed) origin of the –96 mV deficit.

7. References

1. M. Burgelman, P. Nollet, S. Degraeve. “Modelling polycrystalline semiconductor solar cells.” *Thin Solid Films* **361–362**, 527–532 (2000). — Core SCAPS drift-diffusion formulation; Gummel/Newton solver.
2. M. Burgelman, J. Marlein. “Analysis of graded band gap solar cells with SCAPS.” *Thin Solid Films* **515**(15), 6276–6278 (2007). — Intra-band tunnelling at heterointerfaces (OFF in the partner model).
3. K. Decock, S. Khelifi, M. Burgelman. “Modelling multivalent defects in thin-film solar cells.” *Thin Solid Films* **519**(21), 7481–7484 (2011). — Defect/grading algorithms.
4. J. Niemegeers, M. Burgelman. Thermionic-emission interface transport and numerical scheme (1998). — Heterojunction boundary condition.
5. D. L. Scharfetter, H. K. Gummel. “Large-signal analysis of a silicon Read diode oscillator.” *IEEE Trans. Electron Devices* **16**(1), 64–77 (1969). — Exponentially-fitted flux discretization.
6. *SCAPS Manual* (February 2016), University of Gent. — pp. 9–10 (contacts/flatband eq. 1–3), pp. 24/26 (neutral defects; τ usage), pp. 30–31 (interfaces: thermionic emission + Pauwels–Vanhoutte; tunnelling), pp. 39–41 (DD eq. 14–15, Gummel + Newton, FD mesh), p. 31 (Boltzmann default), pp. 36–37 (optics, $R = 0$ default).
7. W. Shockley, W. T. Read. “Statistics of the recombinations of holes and electrons.” *Phys. Rev.* **87**(5), 835–842 (1952). — Bulk SRH theory.
8. R. J. Pauwels, G. Vanhoutte. “The influence of interface state and energy barriers on the efficiency of heterojunction solar cells.” *J. Phys. D: Appl. Phys.* **11**(5), 649 (1978). — Interface-state (both-band-edge) recombination theory.
9. J. Nelson. *The Physics of Solar Cells*. Imperial College Press (2003). — SRH/recombination, diode/ J_0 , device physics reference.
10. P. Calado, A. M. Telford, D. Bryant, et al. “Evidence for ion migration in hybrid perovskite solar cells with minimal hysteresis.” *Nat. Commun.* **7**, 13831 (2016) / Driftfusion. — Open-source perovskite DD reference.
11. N. E. Courtier, J. M. Cave, J. M. Foster, A. B. Walker, G. Richardson. “IonMonger: a free and fast planar perovskite solar cell simulator with coupled ion vacancy and charge carrier dynamics.” *J. Comput. Electron.* **18**, 1435–1449 (2019). — V_{bi} -as-free-parameter convention; TE handling.
12. “Effect of perovskite thickness on electroluminescence and solar cell conversion efficiency.” *J. Phys. Chem. Lett.* (2020). doi:10.1021/acs.jpclett.0c02363. — Measured V_{oc} 1.18 V = 94.4% of SQ at 1.53 eV. (*Mod-erate confidence on the SQ-ceiling value, which varies with bandgap definition.*)
13. K. Tvingstedt, O. Malinkiewicz, A. Baumann, et al. “Radiative efficiency of lead iodide based perovskite solar cells.” *Sci. Rep.* **4**, 6071 (2014). doi:10.1038/srep06071. — Non-radiative V_{oc} deficit / EL reciprocity baseline (150–280 mV).

14. L. M. Pazos-Outón, T. P. Xiao, E. Yablonovitch. “Fundamental efficiency limit of lead iodide perovskite solar cells.” *J. Phys. Chem. Lett.* **9**(7), 1703–1711 (2018). doi:10.1021/acs.jpclett.7b03054. — Radiative $\approx$ Auger limit (< 2 mV).
15. W. E. I. Sha, X. Ren, L. Chen, W. C. H. Choy. “The efficiency limit of $\text{CH}_3\text{NH}_3\text{PbI}_3$ perovskite solar cells.” *Appl. Phys. Lett.* **106**, 221104 (2015). arXiv:1506.09003. — SQ/detailed-balance V_{oc} and PCE ceiling for MAPbI_3 .
16. Mesoporous- TiO_2 /spiro champion (luminescent-perovskite cell): J_{sc} 24.6 mA/cm^2 , V_{oc} 1.16 V, FF 0.73, PCE 20.8%. *Energy Environ. Sci.* (2016); PMC4705040. — Direct n-i-p champion bracket for SCAPS’s 1.1676 V.
17. Low-loss MAPbI_3 stack: V_{oc} 1.21 V at 1.53 eV, certified 23.09% PCE, $\Delta V_{oc,nr} \approx 0.10$ V. *Sci. China Mater.* (2025). doi:10.1007/s40843-025-3457-3. — Physical V_{oc} ceiling for this exact bandgap.
18. Single-crystal MAPbI_3 cells $> 21\%$ PCE, V_{oc} 1.0–1.1 V. *ACS Energy Lett.* **4**(5), 1258 (2019). doi:10.1021/acsenergylett.9b00847. — Typical-device V_{oc} regime (where SolarLab’s 1.072 V sits).

Notation: V_{oc} = open-circuit voltage; J_{sc} = short-circuit current density (mA/cm^2); J_0 = dark-saturation current density (A/cm^2); FF = fill factor; PCE = power-conversion efficiency; $V_T = kT/q \approx 25.85$ mV at 300 K; ΔE_C = conduction-band offset; N_D = donor density (cm^{-3}); N_t = trap density (bulk cm^{-3} / interface cm^{-2}); τ = SRH lifetime; n_i = intrinsic carrier density (cm^{-3}); SG = Scharfetter–Gummel; TE = thermionic emission; QFL = quasi-Fermi level; SRH = Shockley–Read–Hall; SQ = Shockley–Queisser; MOL = method of lines. SolarLab code paths are relative to `perovskite-sim/perovskite_sim/` unless otherwise noted; sweep data from `outputs/scaps_full_{off,on,ifacestate}/summary.json` and `perovskite-sim/outputs/scaps_validation/`.